Machine, Data & Learning
Informed Search

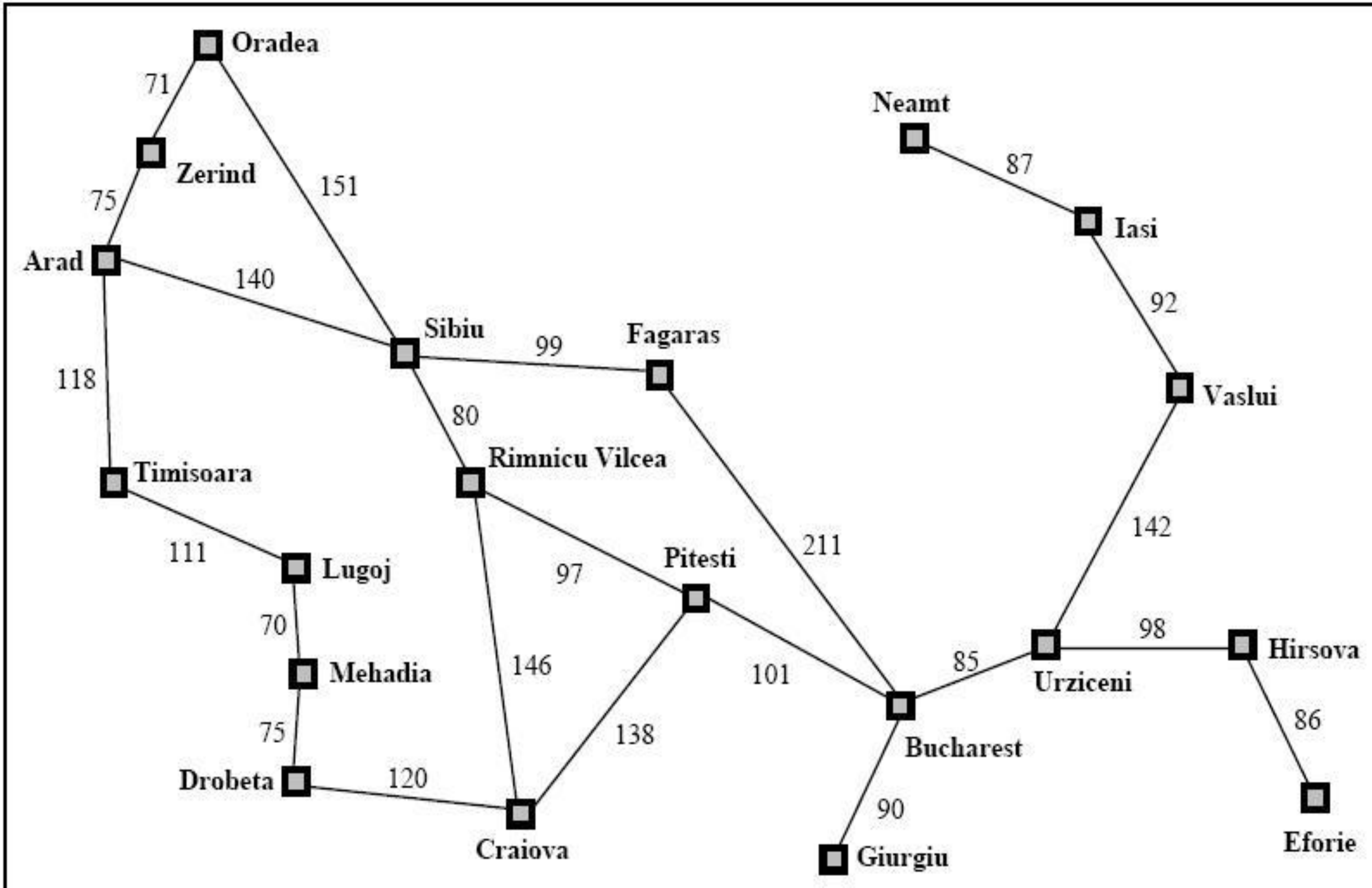
Best First Search
(Chapter 3, Section 5)

- Node selected for expansion based on evaluation function **f(n)**
- Node with lowest cost evaluation expanded first
- Heuristic function **h(n)** as a component of f(n)
- h(n) = *estimated cost of the cheapest path from the state at node n to a goal state*
- **Greedy Best First Search**
- **A* Search**
- **Memory bounded heuristic search**
  - **IDA*, RBFS and SMA***

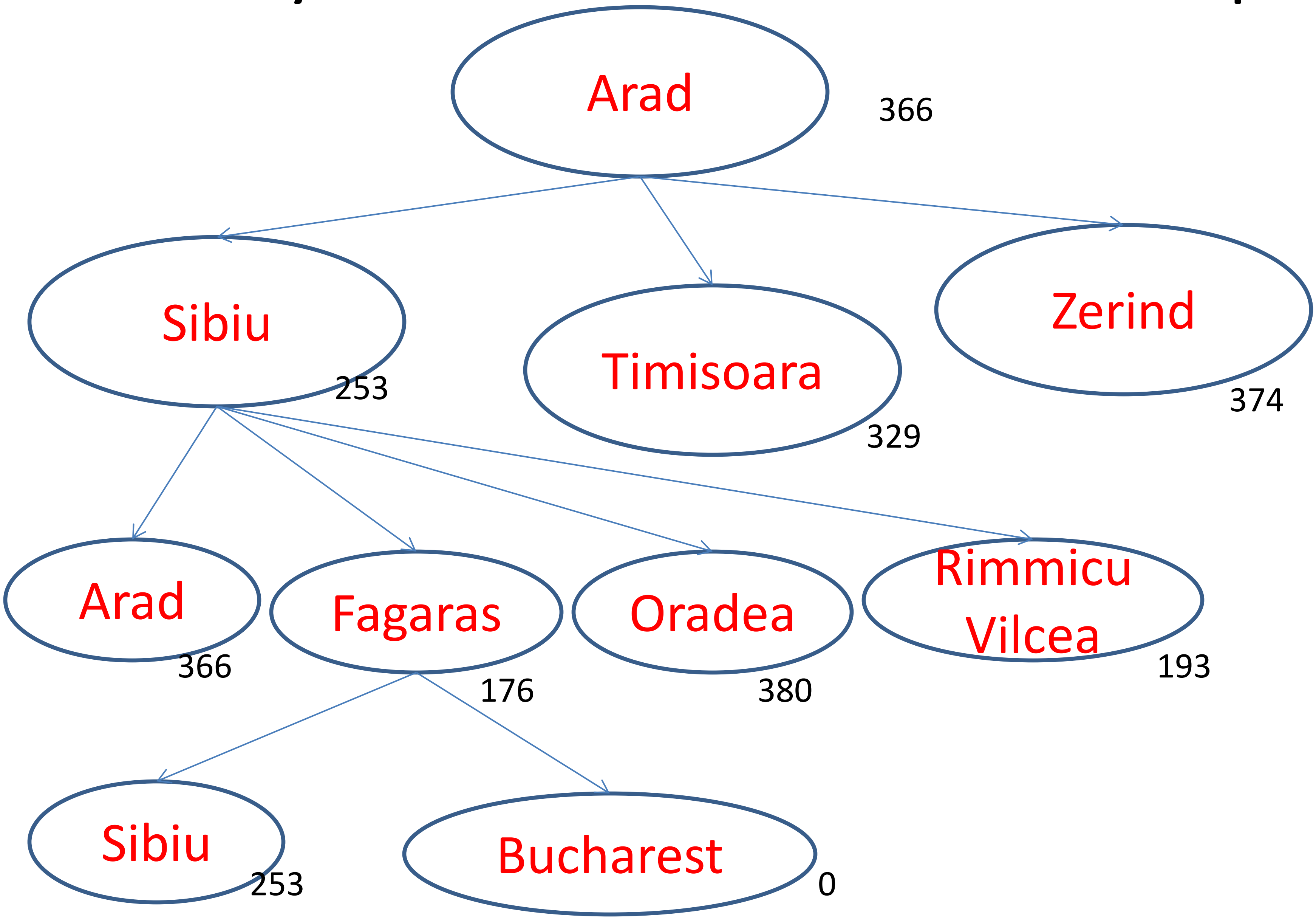
Greedy Best First Search

- Expand node closest to goal
- **f(n) = h(n)**
- For route finding problem straight line distance **h_SLD** can be a heuristic
- h_SLD from Arad to Bucharest is 366 (Next slide)
- **Need not** be optimal
- Tree search is **incomplete** like depth first (infinite loop), graph search is **complete** for finite spaces

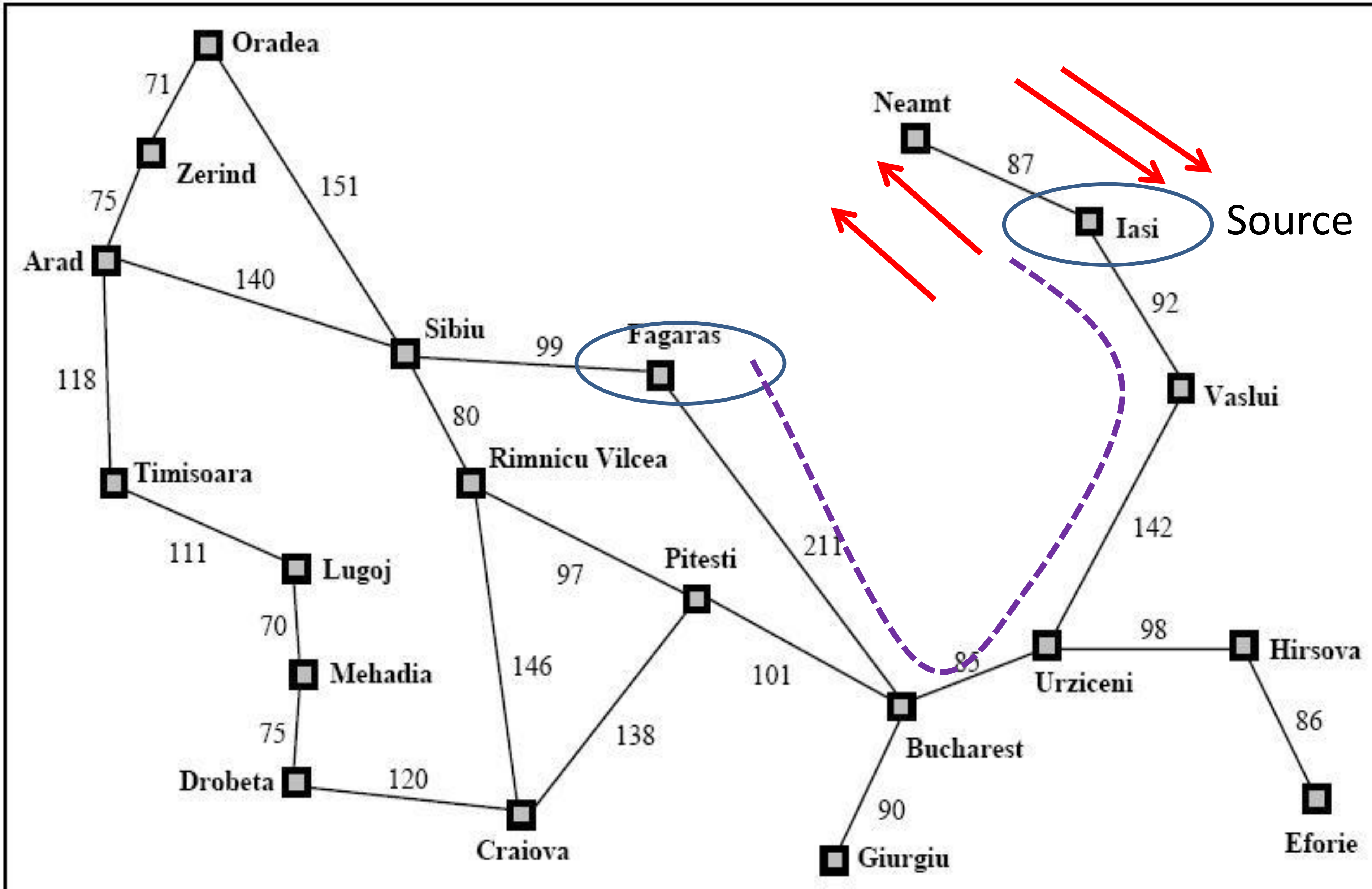
Road map of Romania



Greedy Best First Tree Search Example



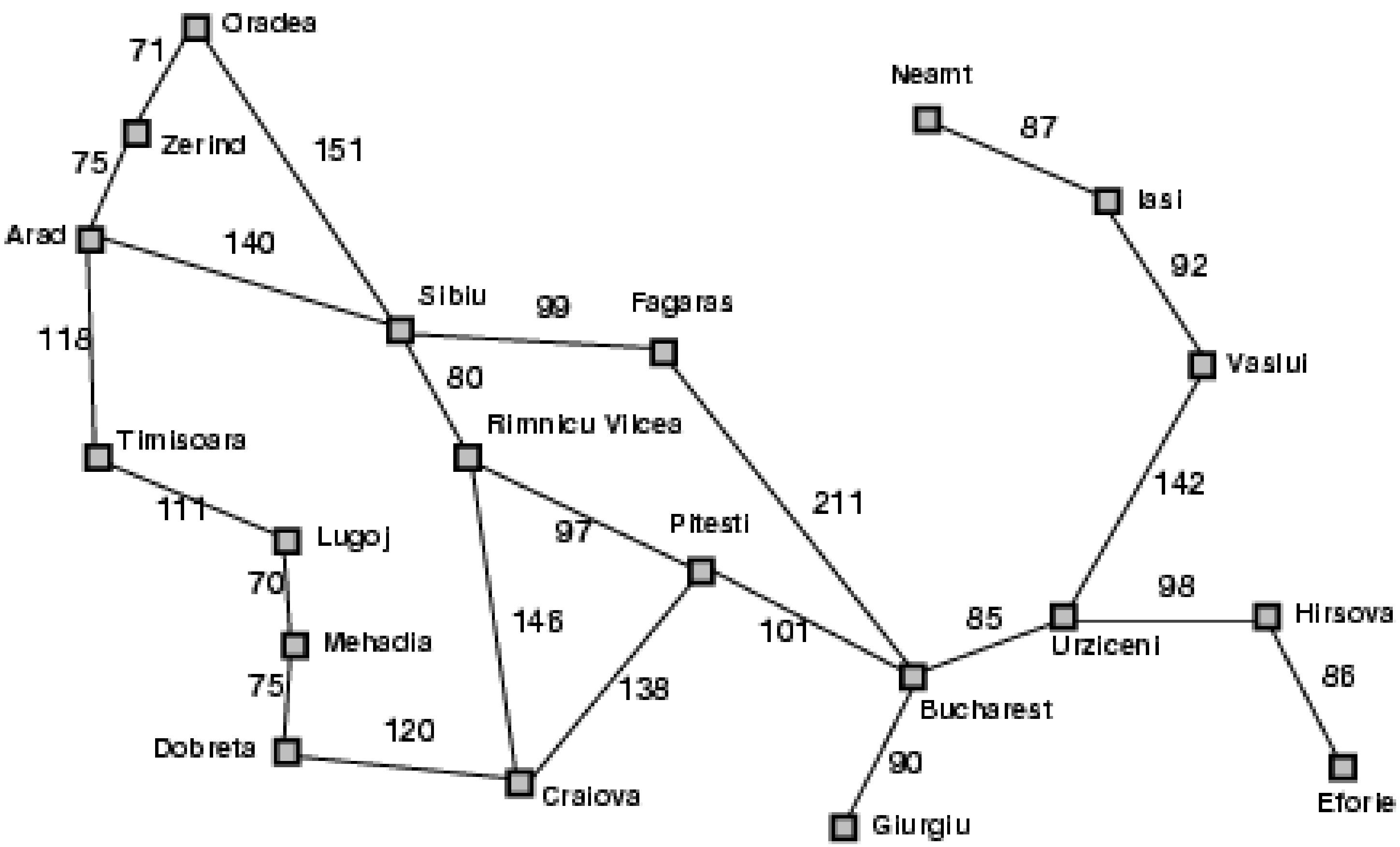
Incompleteness



A* Search

- **f(n) = g(n) + h(n)** i.e. *Cost to reach the node + Cost from node to goal*
- f(n) = *Estimated cost of the cheapest solution through n*
- Expand node with **lowest** f(n)
- If h(n) satisfies certain constraints A* search is complete and optimal
- Algorithm identical to **uniform cost search**

Romania with step costs in km



Straight–line distance to Bucharest

Arad	366
Bucharest	0
Craiova	160
Dobreta	242
Eforie	161
Fagaras	176
Giurgiu	77
Hirsova	151
Iasi	226
Lugoj	244
Mehadia	241
Neamt	234
Oradea	380
Pitesti	10
Rimnicu Vilcea	193
Sibiu	253
Timisoara	329
Urziceni	80
Vaslui	199
Zerind	374

Open List:

Arad

A* search example

We start with our initial state Arad. We take a node and add it to the open list. Since it's the only thing on open list, we expand the node.

Open list sorts the nodes inside of it according to their g()+h() score.

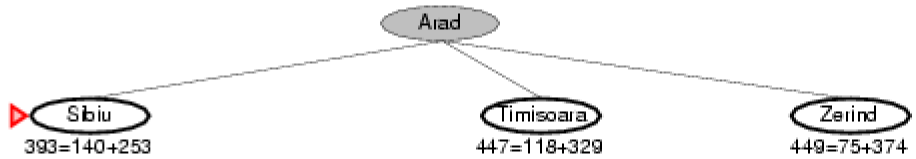
Open List:

Sibiu

Timisoara

Zerind

A* search example



We add the three nodes we found to the open list.

We sort them according to the g()+h() calculation.

Open List:

Rimricu Vicea

Fagaras

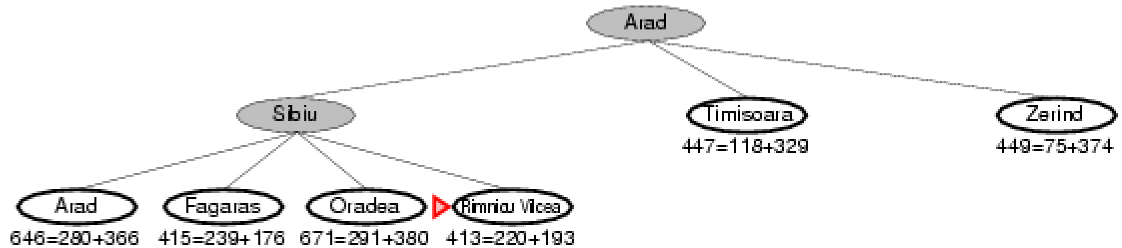
Timisoara

Zerind

~~Arad~~

Oradea

A* search example



We've been to Arad before. Don't list it again on the open list.

When we expand Sibiu, we run into Arad again. But we've already expanded this node once; so, we don't add it to the open list again.

Open List:

Rimricu Vicea

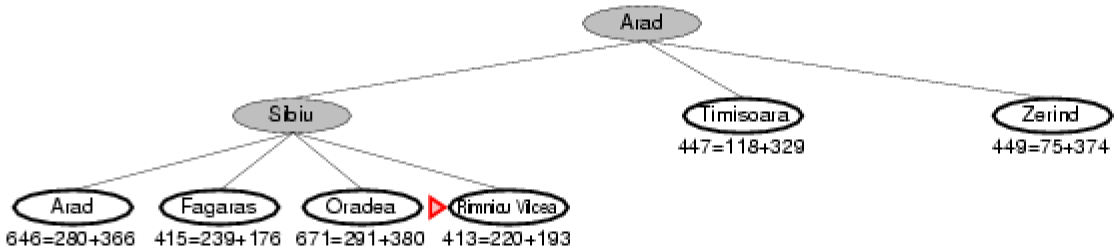
Fagaras

Timisoara

Zerind

Oradea

A* search example



We see that Rimricu Vicea is at the top of the open list; so, it’s the next node we will expand.

Open List:

Fagaras

Pitesti

Timisoara

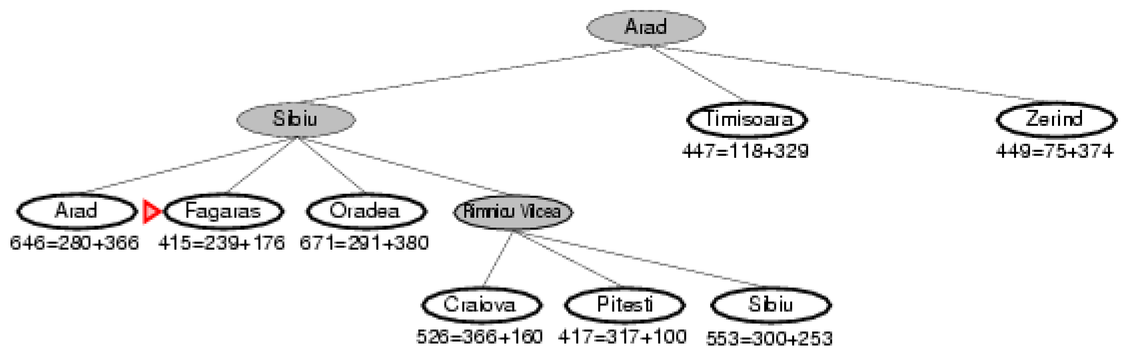
Zerind

Craiova

~~Sibiu~~

Oradea

A^* search example



When we expand Rimricu Vincea, we run into Sibiu again.

But we've already expanded this node once; so, we don't add it to the open list again.

Open List:

Fagaras

Pitesti

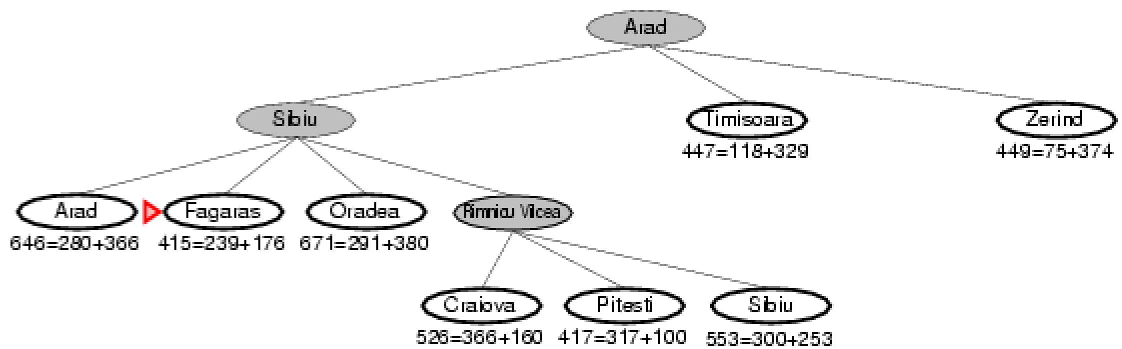
Timisoara

Zerind

Craiova

Oradea

A* search example



Fagaras will be the next node we should expand – it's at the top of the sorted open list.

Open List:

Pitesti

Timisoara

Zerind

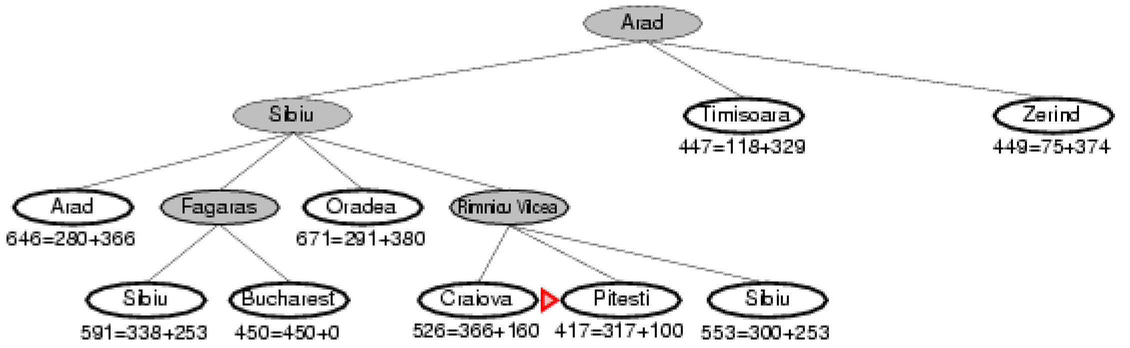
Bucharest

Craiova

~~Sibiu~~

Oradea

A* search example



When we expand Fagaras, we find Sibiu again. We don't add it to the open list.

We also find Bucharest, but we're not done. The algorithm doesn't end until we "expand" the goal node – it has to be at the top of the open list.

Open List:

Pitesti

Timisoara

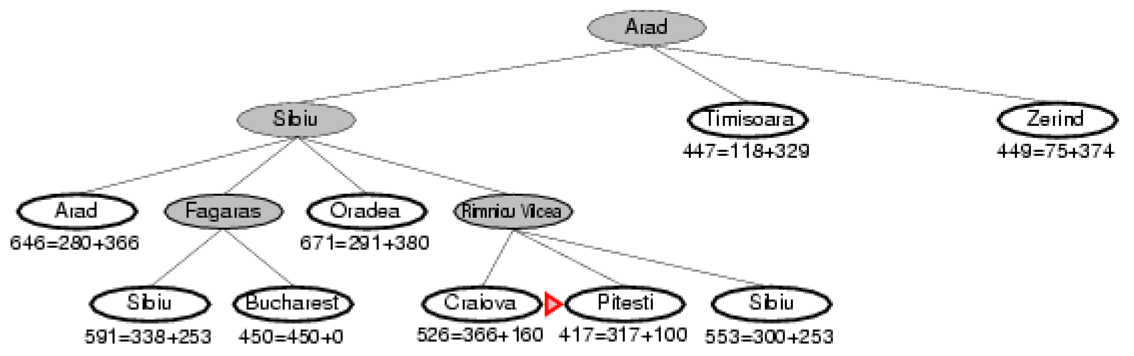
Zerind

Bucharest

Craiova

Oradea

A* search example



It looks like Pitesti is the next node we should expand.

Open List:

Bucharest

Timisoara

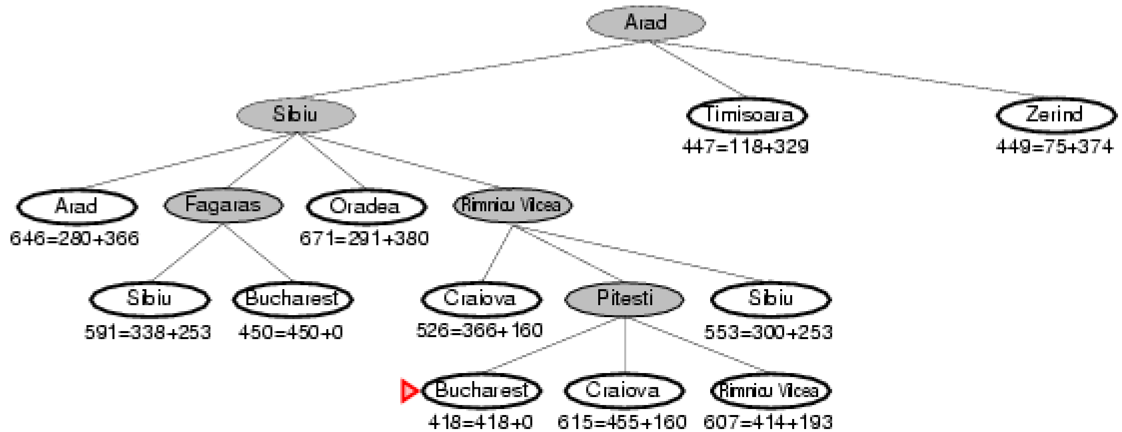
Zerind

Craiova

~~Rimricu Vicea~~

Oradea

A* search example



We just found a better value for Bucharest; so, it got moved higher in the list. We also found a worse value for Craiova – we just ignore this.

And of course, we ran into Rimricu Vicea again. Since it's already been expanded once, we don't re-add it to the Open List.

Open List:

Bucharest

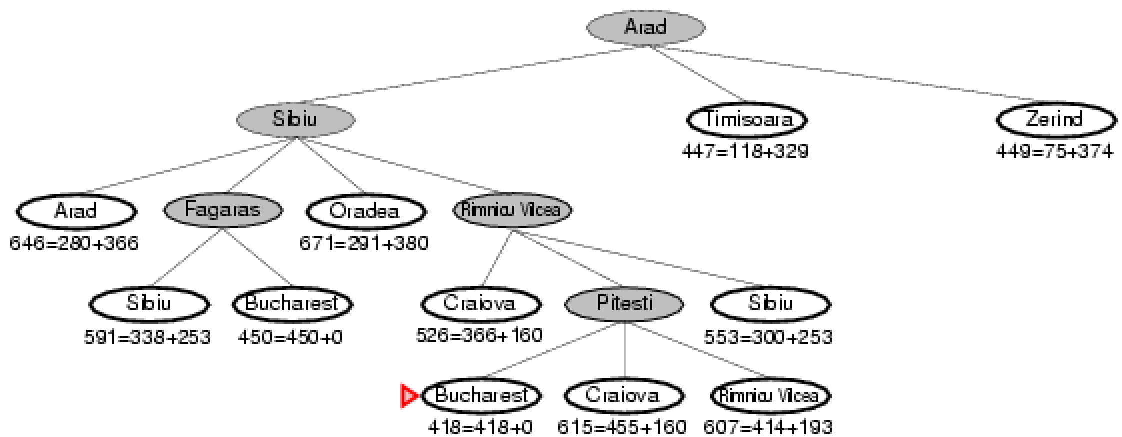
Timisoara

Zerind

Craiova

Oradea

A* search example



Now it looks like Bucharest is at the top of the open list…

Open List:

Bucharest

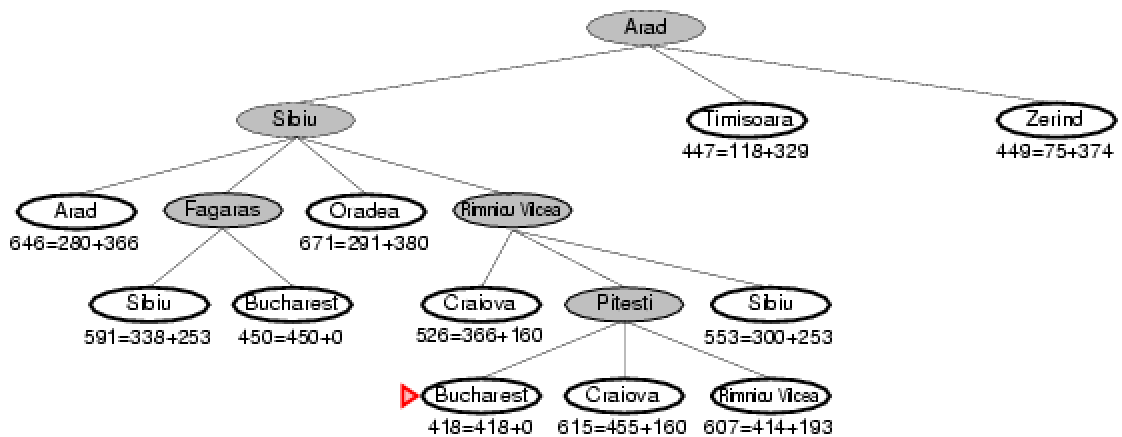
Timisoara

Zerind

Craiova

Oradea

A* search example



Now we “expand” the node for Bucharest.

We’re done! (And we know the path that we’ve found is optimal.)

Conditions for optimality: Admissibility and Consistency

- **Admissibility**: h(n) never overestimates the cost to reach a goal
- Optimistic in nature since h(n) thinks cost of solving problem is lower than it actually is
- Straight line distance h_LSD is admissible
  - Cannot have something shorter than straight line
- **Consistency (or monotonicity)**: Required for graph search
- h(n) <= c(n,a,n') + h(n')
- Form of triangle inequality – Triangle formed by n, n' and goal G_n
- If there is a route from n to G_n via n' cheaper than h(n) it would violate h(n) as lower bound to reach G_n

Optimality of A*

- **Every** consistent heuristic is admissible
- Consistency is a **stricter** requirement although most heuristics are both admissible and consistent
- *Is h_SLD a consistent heuristic ?*
- Tree search version optimal if h(n) is admissible and graph search optimal if h(n) is consistent
- If A* selects a path p, then p is the shortest or lowest cost path

Proof of optimality for A*

- **Step 1:** If h(n) is consistent, then values of f(n) along any path are non decreasing
- **Step 2:** Whenever A* selects a node n for expansion, the optimal path to that node has been found
- **For Step 1**: For definition of consistency, if n' is successor of n
  - $g(n') = g(n) + c(n,a,n')$
  - $f(n') = g(n') + h(n') = g(n) + c(n,a,n') + h(n') >= g(n) + h(n) = f(n) \Rightarrow f(n') >= f(n)$

Proof of optimality for A*

- **For Step 2**: If optimal path to node n was not found before expanding
  - Another frontier node n’ would be on the optimal path from start node to n
  - By the graph separation property (frontier separates the explored region of the state space from the unexplored region), since f is non-decreasing n’ will have lower cost than n and hence would be selected first
- From steps 1 + 2, sequence of nodes expanded is in **non decreasing** order of f(n)
- Hence the **first goal** node selected must be optimal and later goal nodes will be at least as expensive

A* properties

- A* expands all nodes with f(n) < C*
- A* might expand nodes on goal contour (f(n) = C*) before selecting a goal node
- A* does not expand nodes with f(n) > C*
- A* is complete i.e. will always find a solution
- A* is optimally efficient for any given consistent heuristic
  - No other optimal algorithm is guaranteed to expand fewer nodes than A* given the heuristic
  - If an algorithm does not expand all nodes with f(n) < C*, it may miss the optimal solution
- Hence A* is **complete, optimal and optimally efficient**

A* properties

- Complexity of A* dependent on quality of heuristic function chosen, varies with assumptions made on state space
- **Absolute error Δ** ≡ h*- h (h* is actual cost of getting from root to goal, h the estimated cost)
- **Relative error Є** ≡ (h*-h)/h*
- For tree with reversible actions and single goal, O(b^Δ)
  - With constant step costs, O(b^ Єd) = O((b^ Є)^d)
- With multiple goal states
  - Extra cost proportional to number of goals (whose cost is within factor Є of optimal cost)
- For general graph, exponentially many states with f(n) < C*
- **While computationally expensive to find optimal solutions, usually runs out of space before time**